

SuperResearch · Guided Tour

SuperResearch, a real deep-research harness, unlike essay-writing models.

A screen-by-screen tour: every setting explained, every stage in the order you'll meet it, and why each piece exists.

~12 min read Beginner friendly v0.2

Six stages, start to finish

SuperResearcher runs on your own computer. You describe a topic, and it does the legwork: finding real sources, reading them, organizing what they say, and writing it up as a polished report with charts pulled from the sources themselves. One page of this guide per stage, in the order you'll click them:

1 Scope Describe the research and set the dials. The run executes here.	2 Map Explore the embedding Atlas of everything it read.	3 Curate Shape the topic tree: keep what matters, cut the rest.	4 Compose Assemble the evidence each section will cite.	5 Compile Draft the outline, then write the paper section by section.	6 Publish Review figures, pick a template, export PDF & HTML.
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How to read this guide: each of the next six pages covers one stage — what the screen shows, what each control means and why it exists, and the one action that moves you forward.

Scope: the research brief

Every control on the form, what it means, and why it's there. The topic is the only required field — the rest tune how the run behaves.

Research topic Required

The subject in plain language. Drives queries, source selection, topics, chapters.

Why: the app's only definition of "relevant." Specific beats clever.

Depth

Search results considered **per query**.

Low 3

Med 7

High 10

X-High 15

Ludicrous 50

Why: the thoroughness dial. Richer corpus, longer run.

Breadth

Angles of attack generated per topic when expanding queries.

Low 1

Med 3

High 5

Why: depth digs one query; breadth covers angles you wouldn't think of.

Final sources

Target corpus size. Discovery stops here (default **120**, max **500**).

Why: caps the run — every source is downloaded, chunked, embedded.

Research context

Why it matters, who it's for, what decision it supports.

Why: an investor's report reads differently from a founder's. Context picks which.

Scope controls (advanced)

Audience, geography, time horizon, objective, must include / exclude.

Why: guardrails — "must exclude: consumer drones" keeps the report on rails.

AFTER START — RUN PHASES

Dossier → folder

Protocol → plan

Heuristics → queries

Discover → candidates

Select → final set

Ingest → download & extract

API KEYS · REQUIRED

EXA_API_KEY=exa_...

search — no key, no discovery

GEMINI_API_KEY=AIza...

writing — quality comes from this model

in api_keys.txt, never committed

Map: the embedding Atlas

Every passage becomes an **embedding** — a numeric fingerprint of what it *means* — plotted on a 2D map. Dots together say similar things. **Why embeddings, not keywords:** keywords find only words you thought of; embeddings find *meaning* — "battery energy density" lands beside "cell specific energy" with zero shared words. Topic discovery sees the real themes, and every citation is retrieved by meaning.



Coverage at a glance

Tight clusters mean deep coverage; empty regions mean angles the corpus barely touches — a cue to run discovery again.



Evidence on demand

Click any dot to read the exact passage — source, page, chunk ID. Every claim in the report traces back here.



Themes become chapters

Clusters turn into the topic tree on the next page — the map is where the report's structure comes from.

Curate: shape the topic tree

The Atlas clusters become a **topic tree** — the chapters and subtopics your report could cover. This is your **editorial pass**: **Keep** what matters, **Reject** what doesn't. Nothing rejected here reaches the report. The tree is discovered from your corpus, not invented — every branch traces back to real clusters of passages.

http://127.0.0.1:8765 – Curate · 10 chapters, 55 subtopics

YOUR CALL

1. eVTOL aircraft designs **Keep**

· Vectored thrust **Keep**

· Multirotor configurations **Keep**

· Wingless concepts **Reject**

2. Certification pathways **Keep**

· FAA type certification **Keep**

· EASA SC-VTOL **Keep**

· Military airworthiness **Reject**

**Be ruthless**
Ten minutes here shapes the entire report. A tight tree beats a sprawling one.

**Rejected stays rejected**
Cut branches never come back — the report can't drift into territory you ruled out.

**Cut too deep?**
Re-run topic discovery any time. Curation is reversible until you compile.

Compose: brief the writer

For each section, you prepare **query terms** — the phrases the app uses to pull the right evidence from the Atlas at writing time. **Why this stage exists:** writing straight from 3,000 chunks would drown the model in irrelevant passages. Compose narrows each section to a focused **evidence bundle** — a few dozen top chunks — so every section is written from the best material available, and every claim carries a real citation. Finalize the terms here; Compile builds on exactly this.

http://127.0.0.1:8765 — Compose · query-term preparation

SECTION 4: VERTIPORT INFRASTRUCTURE — COMPOSE TERMS

vertiport siting requirements · megawatt charging infrastructure · urban air traffic management · ground footprint per pad · downtown skyport proposals

EVIDENCE BUNDLE

52 chunks from 11 sources

top match S0014:C0087

coverage: strong · recency: strong



Precision over volume

46 hand-checked chunks beat 3,000 unfiltered ones. The bundle is what the writer actually sees.



Thin bundle?

A weak bundle means weak terms — refine them here. Far cheaper than rewriting the section later.



Lock before compiling

Finalize every section's terms first. Compile reads these bundles and nothing else.

Compile: write the paper

Two moves. First **Plan TOC** — the app drafts a chapter-by-chapter outline from your curated topics. Read it like an editor; tick *Rebuild TOC* and run it again until the shape is right. Then **Compile paper** writes every section one by one, each from its evidence bundle, each factual claim carrying an inline citation. The outline is a draft, not a commitment — rebuild as many times as you like before a word is written.

http://127.0.0.1:8765 – Compile · section 23 of 55

COMPILE PAPER

TABLE OF CONTENTS · DRAFT

1. The eVTOL landscape: players and promises

2. Aircraft designs and trade-offs

3. Batteries: the binding constraint

4. Vertiports: the ground game

... 51 more sections

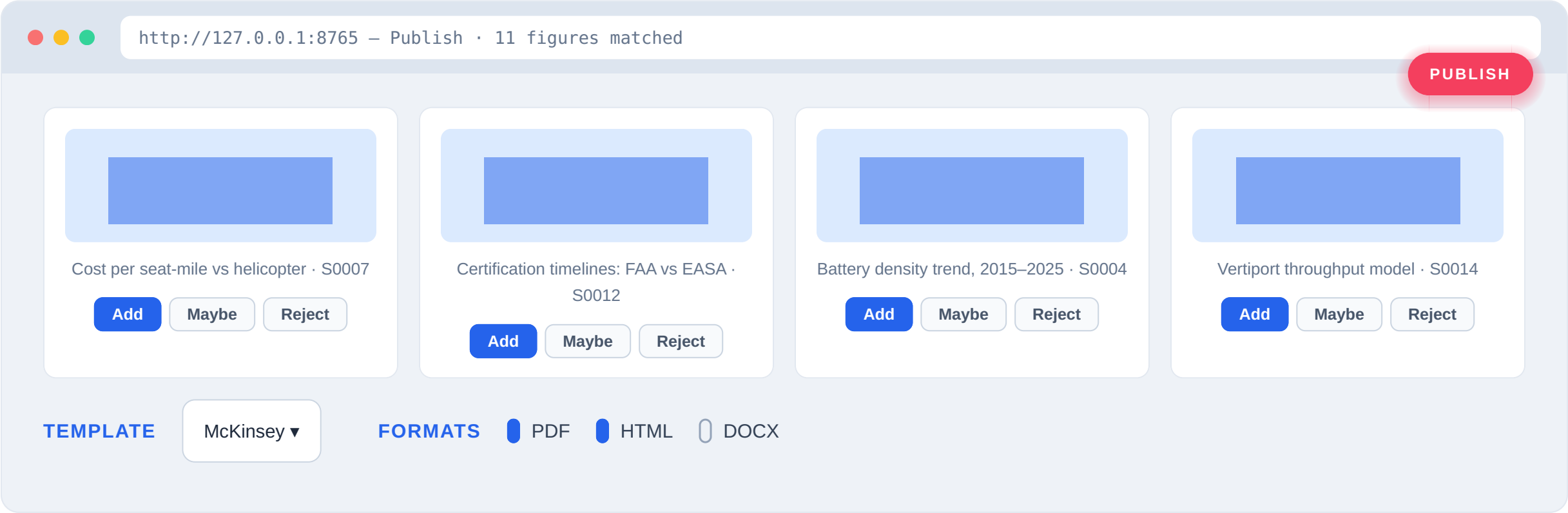
LATEST SECTION

Pack prices fell below \$120/kWh for the third straight year S0004:C0117 , yet usable energy density gains have slowed to roughly 4% annually S0007:C0142 ...

Reading a citation: S0001:C0107 means source 1, chunk 107. Open the Map tab's chunk inspector to read the exact passage behind any sentence in the report.

Publish: figures, then export

Two jobs. **Visual review** first: the app pulls charts and figures straight out of your source PDFs and matches each to the section it belongs in — **Add** the good ones, **Reject** the duds. These are the real charts from the real sources, each captioned with where it came from. Then **export**: pick a template, tick your formats, hit Publish.



Done: finished files land in the run folder under `atlas/publish/` — open the PDF first, then the HTML to copy text or links. Two templates: **McKinsey** and **Meta Research**.

Questions, answered

The things first-time users usually ask.

How long does a full report take?

Under an hour for a focused brief; several hours for a 50-section deep dive over 100+ sources. Live progress throughout — close the browser and come back anytime.

Which API keys do I need?

A search key (Exa or Serper) and a language-model key (Gemini, or the Codex CLI) — in `api_keys.txt` before your first real run.

Can I see a finished report first?

Yes — the repo's `outputs/` folder has complete reports, including the eVTOL air-mobility deep dive (55 sections, 11 source figures).

Where do my reports live?

`research_runs/<date-topic>_Corpus/` — originals, sidecars, the Atlas, finished PDFs. Nothing locked inside the app.

What if something breaks halfway?

Each stage resumes where it stopped — completed work is never redone. Final verdicts are honest: Pass, Pass with warnings, or Fail.

Do I need to babysit it?

No. Set the brief, hit Start, walk away. The two moments that want you are Curate (your editorial pass) and Publish (figure review).